

LESION-LEVEL SYNTHETIC DATA GENERATION FOR CO-EXISTING RICE LEAF DISEASE RECOGNITION

Abstract—The availability of rice leaf disease datasets containing co-existing diseases is limited, as most publicly available datasets primarily consist of single-disease images. This limitation restricts the development and evaluation of multi-label disease recognition models for realistic agricultural conditions. This work proposes a lesion-level synthetic dataset generation pipeline for creating rice leaf images containing co-existing diseases from curated single-disease datasets. The proposed framework performs dataset cleaning, image preprocessing, lesion segmentation using Segment Anything Model (SAM)[26], lesion extraction, and controlled lesion transfer between rice leaves. Feathered alpha compositing and multi-band Laplacian pyramid blending[9] are employed to reduce boundary artifacts and preserve visual continuity between transferred lesions and target leaves. Synthetic samples representing BLB+Blast, BLB+Brown Spot, and Blast+Brown Spot combinations are generated and subsequently augmented to obtain a balanced dataset. The generated images are evaluated using PSNR, SSIM, NIQE, BRISQUE, FID, visual inspection, and agriculture-domain expert assessment. The resulting dataset was further evaluated using SVM[20], EfficientNet-B4[23], ViT-Base/16[17], and Swin Transformer models[22] for multi-label disease classification. ViT-Base/16 and Swin Transformer achieved accuracies of 87.68% and 88.41%, respectively, with Swin Transformer obtaining an F1-score of 94.06%. The results demonstrate that controlled lesion-level synthesis can provide a consistent and useful approach for constructing co-existing rice disease datasets.

I. KEYWORDS

Rice leaf disease; Synthetic dataset generation; Co-existing disease detection; Lesion-level image synthesis; Segment Anything Model (SAM); Multi-band Laplacian Pyramid Blending; Multi-label classification; Vision Transformer (ViT).

II. INTRODUCTION

Rice is one of the world's most important staple crops, serving as the primary food source for more than half of the global population. Therefore, ensuring healthy rice production is essential for food security and sustainable agriculture. However, rice cultivation is frequently affected by diseases such as **Bacterial leaf Blight (BLB)**, **Blast**, and **Brown Spot**, which significantly reduce crop yield and quality, leading to considerable economic losses. Consequently, the early and accurate identification of rice diseases has become an important research area in precision agriculture.

Recent advances in machine learning (ML) and deep learning (DL) have enabled automated rice disease diagnosis using leaf image datasets. However, most publicly available datasets contain single-disease images, whereas multiple diseases may

coexist on the same leaf in real agricultural environments. This limits the development and evaluation of multi-label disease recognition systems. Domain generalization has also been explored to improve plant disease recognition across different environments [13].

Co-existing disease datasets are difficult to construct because naturally co-occurring samples are scarce and require labor-intensive expert annotation. Conventional augmentation can increase dataset diversity but does not realistically introduce multiple disease patterns on a single leaf.

To address this limitation, this work proposes a synthetic dataset creation pipeline for generating realistic coexisting rice leaf disease images from curated single disease datasets. The proposed framework combines lesion extraction, image construction and comprehensive validation to produce biologically plausible synthetic samples suitable for downstream machine learning and deep learning applications.

The main contributions of this work are:

- A lesion-level pipeline for generating synthetic coexisting rice leaf disease images from curated single-disease datasets.
- A synthesis and blending framework for constructing realistic multi-disease leaf images.
- A validation framework combining objective image-quality metrics and agriculture-domain expert assessment.
- A balanced synthetic dataset for multi-label rice disease recognition.

III. RELATED WORK

- 1) **Rice Leaf Disease Datasets** Publicly available rice leaf disease datasets have played a major role in developing automated disease diagnosis systems. Datasets such as **Paddy doctor**[1] and **Dhan-Shomadhan**[2] provide annotated images of major rice diseases, including Bacterial Leaf Blight (BLB), Blast and Brown Spot, and have been widely used for benchmarking ML and DL models. However, these datasets primarily contain single disease leaf images, making them unsuitable for studying coexisting diseases and multi-label classification.
- 2) **Multi-Disease Learning and Synthetic Data Generation** Recent studies have explored multi-label and multi-output disease recognition using architectures such as Vision Transformers, EfficientNet-B4[23], and multi-task learning[3],[18]. CNN-based transfer learning approaches using architectures such as EfficientNet have

also been investigated for rice leaf disease classification [15]. In parallel, synthetic data generation techniques including **copy-paste augmentation** [7][10], Poisson image editing, image harmonization, and **GAN-based augmentation**[8],[19] have been investigated to increase dataset diversity[14]. While these approaches improve model performance, they have seen limited application in generating realistic coexisting rice leaf disease images. Diffusion-based generative models[11] have recently emerged as an alternative to GAN-based approaches for plant disease image augmentation, offering improved realism and diversity in synthesized disease images

- 3) Image Preprocessing and Lesion Extraction, Image Preprocessing is an essential step in plant disease analysis. Techniques such as background removal [4], CLAHE[5], Gaussian filtering, adaptive thresholding, contour extraction, and connected component analysis have been widely adopted to improve lesion visibility and isolate diseased regions. These methods provide a strong foundation for accurate lesion extraction and synthetic image generation. Recent studies have also investigated synthetic plant disease image generation to improve downstream segmentation and recognition tasks [12].
- 4) Research Gap Despite significant progress in disease classification and synthetic image generation, publicly available rice datasets remain limited to single disease samples. Existing augmentation methods generally focus on increasing data diversity rather than creating biologically realistic disease images at the lesion level. Furthermore, comprehensive validation that combines objective image quality metrics with expert agricultural assessment is rarely reported. To address these limitations, this work proposes a lesion-level synthetic dataset generation approach supported by both quantitative evaluation and expert validation.

IV. PROPOSED METHODOLOGY

This work presents a systematic pipeline for generating a synthetic dataset of coexisting rice leaf diseases through lesion-level image synthesis. The proposed methodology comprises dataset collection, preprocessing, lesion extraction, synthetic image generation, data augmentation[7], and validation. The primary objective is to create biologically plausible multi-disease rice leaf images that can support the development of robust multi-label disease recognition models. The overall workflow of the proposed framework is illustrated in Fig.1.

A. Dataset Collection and Preparation

The initial stage involved collecting publicly available rice leaf disease datasets containing images of Bacterial Leaf Blight(BLB), Blast and Brown Spot. Multiple datasets were combined to improve disease diversity and reduce dataset-specific bias. A systematic data-cleaning procedure was then performed to obtain a reliable set of single-disease rice leaf images for subsequent lesion-level synthesis. The cleaning process included image integrity verification, removal of exact

duplicates using **MD5 hashing** and identification of very similar images using **perceptual hashing(pHash)**. Additional quality screening was performed to identify images affected by severe blur, poor exposure and complex field-level backgrounds. Images identified through these automated quality checks were further inspected manually where visually unclear or unsuitable images were removed to maintain consistency and reliability of the dataset. Images containing multiple rice leaves, healthy leaves, or poor visual quality were excluded to maintain consistency. The remaining images were standardized to form a curated single-leaf dataset suitable for lesion-level image synthesis.

B. Image preprocessing

The curated dataset was preprocessed to enhance lesion visibility and improve the quality of subsequent segmentation. **Contrast Limited Adaptive Histogram Equalization (CLAHE)** [5] was applied to the L-channel of the LAB color space to enhance local contrast without amplifying image noise, as shown in Fig.2. The CLAHE operation was configured with a clip limit of 1.8 and an 8×8 tile grid. Following contrast enhancement, Gaussian filtering using a 3×3 kernel was applied to suppress minor artifacts while retaining visible lesion structures, as shown in Fig.3. introduced during contrast enhancement while retaining the visible lesion structures. Background removal [4] was then performed to isolate individual rice leaves from complex backgrounds, enabling accurate lesion extraction and reducing interference during the synthesis process.

C. Lesion Segmentation and Extraction

Disease lesions were segmented using the **Segment Anything Model (SAM)**[26] with a ViT-H backbone. The SAM model was loaded from a pretrained checkpoint and applied to the curated single-disease rice leaf images. To guide segmentation toward disease-affected regions, bounding boxes were manually drawn around the visible lesion regions through an interactive annotation interface. Multiple bounding boxes could be provided for an image when required, and SAM generated multiple candidate masks for each box. The mask with the highest prediction score was selected. The corresponding lesion extraction stage matched each saved mask with its original image and converted the mask into a binary representation. The resulting lesion patches and their corresponding masks were retained as inputs to the synthetic coexisting-disease image generation stage. Representative examples of the segmentation and extraction process are shown in Fig. 2.

D. Synthetic Image Generation

The proposed synthetic image generation framework creates coexisting disease images by transferring lesions from a source leaf to a target belonging to a different disease category. Initially, the extracted lesion patches were positioned at appropriate locations on the target leaf using **feathered alpha compositing**, providing smooth transitions between the pasted lesion and the surrounding tissue. To further improve visual realism and reduce boundary artifacts, **Multi-band Laplacian Pyramid blending**[9] was employed. This blending technique preserves texture continuity across multiple frequency bands

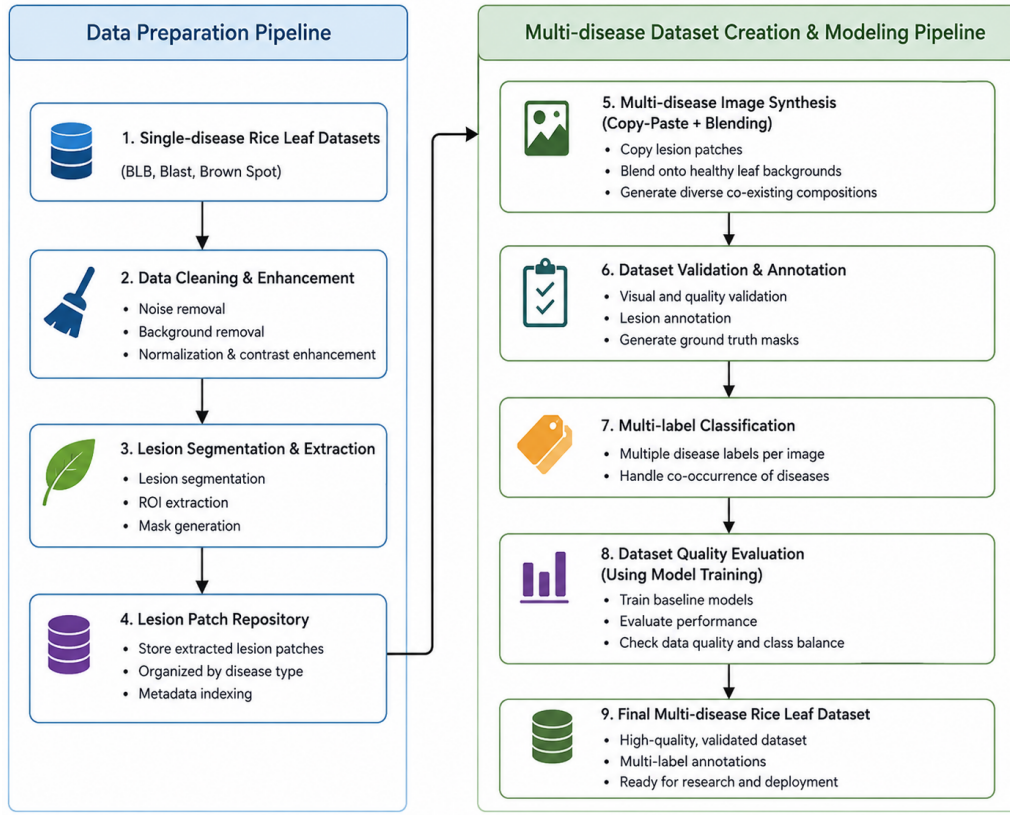


Fig. 1: Overall workflow of the proposed pipeline for multi-disease rice leaf dataset creation, validation, classification, and quality evaluation.

while maintaining the structural characteristics of both the lesion and the target leaf. Using this process, realistic coexisting disease combinations such as **BLB+blast**, **BLB+brown Spot**, and **Blast+Brown Spot** were generated as shown in Fig 5.

E. Data Augmentation

Following synthetic image generation, data augmentation [7] was performed to increase dataset diversity and achieve class balance. Geometric transformations, including rotation, horizontal and vertical flipping, scaling, and brightness adjustment, were applied while preserving disease morphology and lesion characteristics. The augmentation process expanded the dataset and improved its suitability for training ML, DL, and Vision Transformer-based classification models.

F. Dataset Validation

To ensure the quality and reliability of the generated dataset, both quantitative and qualitative validation strategies were employed. Quantitative evaluation was conducted using image quality assessment metrics, including **Peak Signal-to-Noise Ratio (PSNR)**, **Structural Similarity Index (SSIM)**[25], **Natural Image Quality Evaluator (NIQE)**, **Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE)**, and **Frechet Inception Distance (FID)**. These metrics were used to evaluate structural similarity, perceptual quality, and the distributional consistency of the synthesized images. The validation notebook successfully matched all evaluated synthetic images with their corresponding base images, and Both metrics

were computed to quantify the structural similarity between the synthesized images and their corresponding reference images. Boundary artifacts were additionally examined for synthetic images whose base-leaf correspondence could be reliably established from the filename identifiers. For these matched samples, an approximate lesion mask was derived from the pixel-wise difference between the original and synthetic images, followed by Otsu thresholding [6] and morphological processing. A boundary-artifact score was then calculated using color, gradient, and Laplacian differences around the estimated lesion boundary, allowing samples with relatively higher boundary discontinuities to be identified for visual inspection. To assess overall image naturalness without requiring a reference image, NIQE and BRISQUE scores were computed for the synthetic images, with lower values indicating greater perceived naturalness according to these metrics.

In addition to objective evaluation, the generated images were reviewed by an **agriculture domain expert** to assess their biological plausibility.

G. Model Training

To check the applicability of the generated synthetic dataset for automated rice leaf disease recognition, four representative models were trained, A **Support Vector Machine(SVM)**[20], **EfficientNet-B4** [23], **Swin Transformer** [22] and **ViT-Base/16** [21] were selected to represent traditional machine

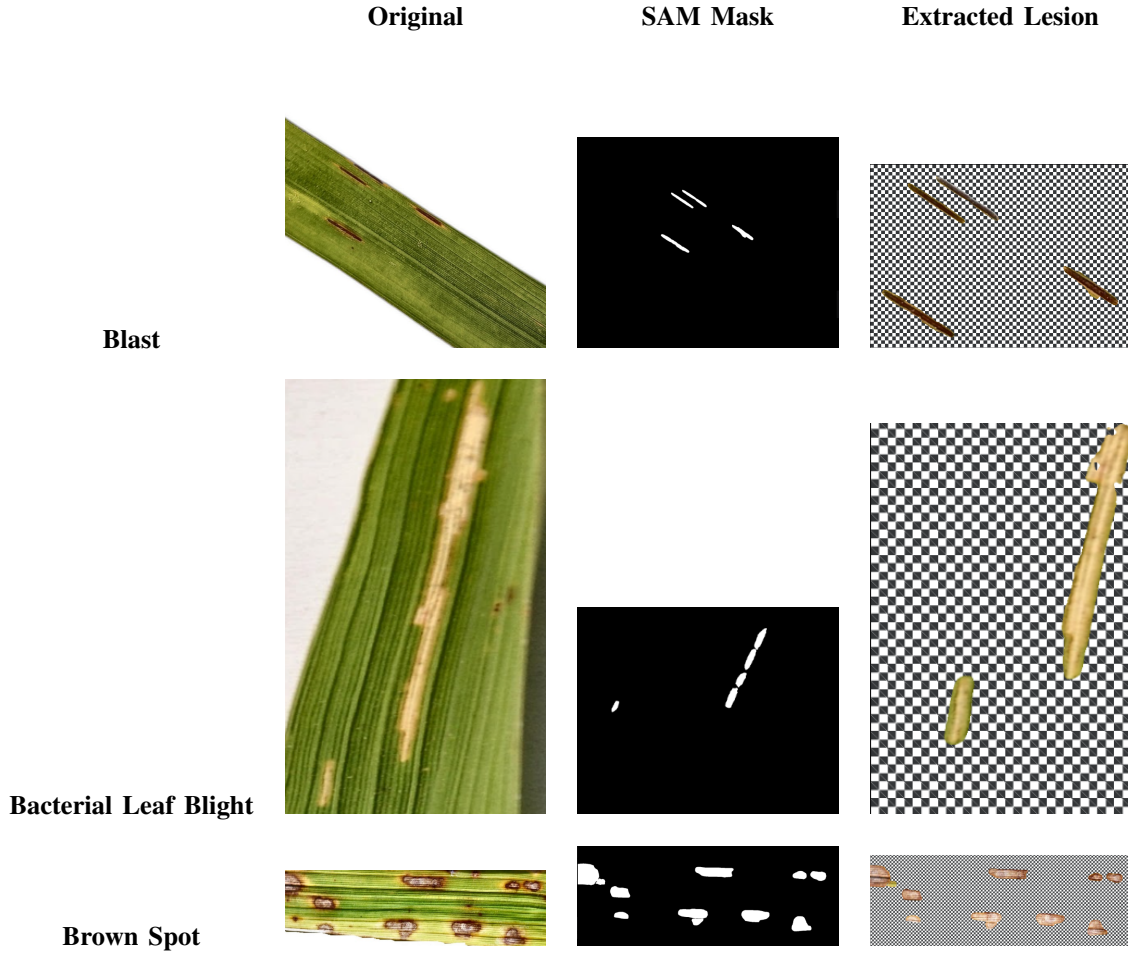


Fig. 2: Representative examples of lesion segmentation and extraction for Blast, Bacterial Leaf Blight and Brown Spot. Each row shows the original diseased leaf, SAM-generated lesion mask, and extracted lesion patch.

learning, CNN, and transformer-based approaches, respectively.

The generated dataset was used to train and evaluate these models under a consistent experimental protocol. Classification performance was assessed using standard evaluation metrics, including **Accuracy, Precision, Recall, and F1-score**. The results obtained from these models provide an indication of the effectiveness of the proposed synthetic dataset in supporting multi-label rice leaf disease classification.

V. EXPERIMENTAL SETUP

This section describes the experimental environment, dataset composition, evaluation metrics, and model configurations used to assess the quality and effectiveness of the proposed synthetic rice leaf disease dataset.

1) Experimental Environment

All experiments were conducted using Python 3.x in the Google Colaboratory environment. The proposed framework was implemented using **OpenCV, NumPy, Pandas, PyTorch, and Scikit-learn** libraries. Image preprocessing and synthetic image generation were performed using OpenCV-based image processing tech-

niques. Lesion extraction was done using the **Segment Anything Model (SAM)**[26]. The experiments were executed on an NVIDIA Tesla T4 GPU with 15 GB GPU memory. This provides sufficient computational resources for segmentation, image synthesis, and model training.

2) Dataset Description

The proposed dataset was created using publicly available Kaggle rice leaf disease datasets namely Paddy Doctor [1] and Dhan-Shomadhan [2]. These datasets contain images of Bacterial Leaf Blight(BLB), Blast, and Brown Spot. After dataset cleaning and quality assessment, only high quality single-leaf disease images were retained for synthetic image generation.

Lesion-level synthesis was performed to generate three coexisting disease combinations:

- A. BLB + Blast
- B. BLB + Brown Spot
- C. Blast + Brown spot

To address class imbalance and improve dataset diversity, the synthesized images were further augmented

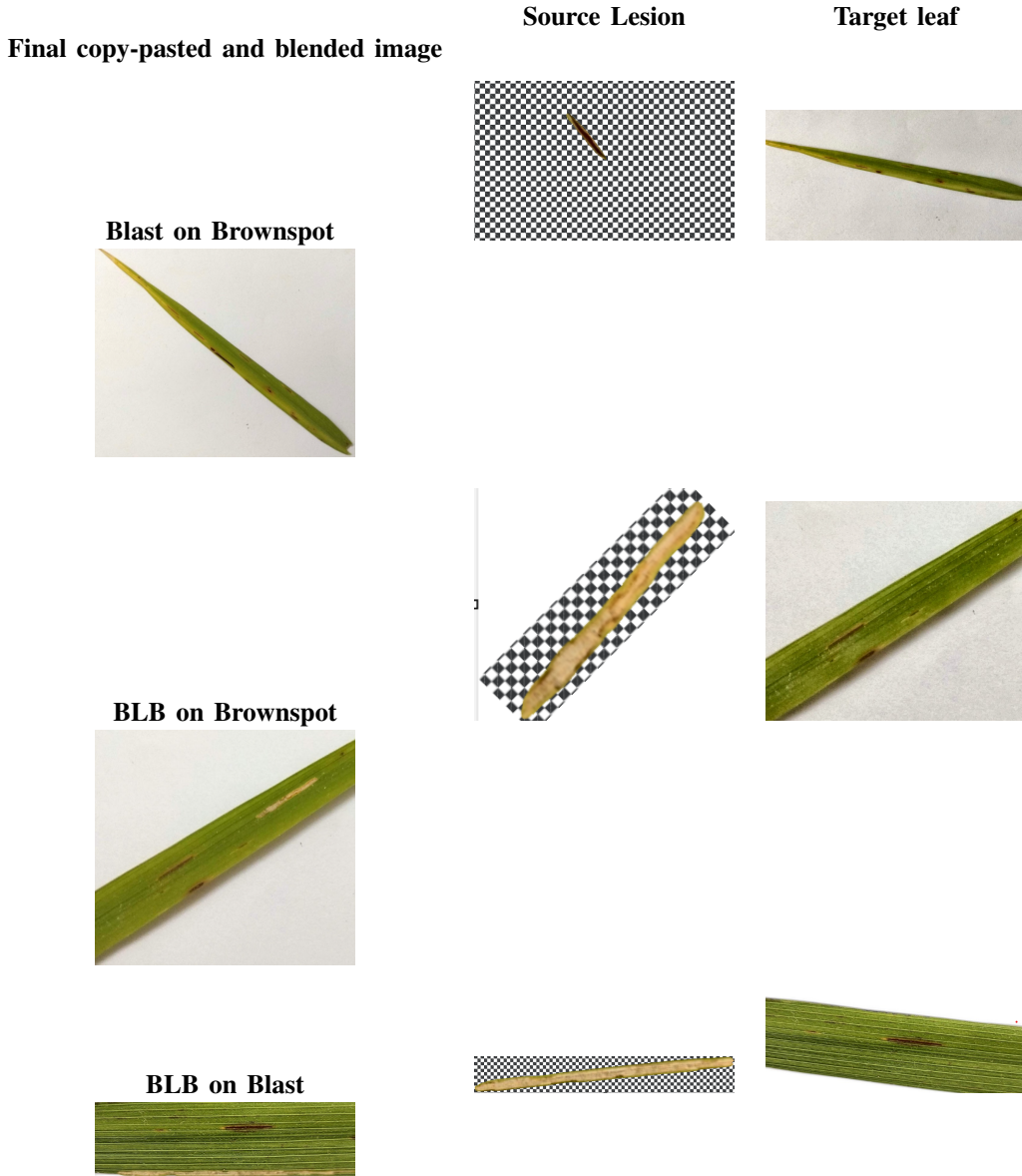


Fig. 3: Lesion-level synthesis of co-existing rice leaf disease images using lesion transfer and multi-band Laplacian pyramid blending.

using geometric and photometric transformations. The final dataset consisted of balanced classes suitable for training and evaluating ML, DL, and ViT models. Dataset Statistics are in **Table I** and **Table II**. Although a larger number of candidate images were produced during copy-paste synthesis, a subset was retained after blending and quality screening. Samples exhibiting severe boundary artifacts, inappropriate lesion placement, or visually inconsistent compositions were excluded. The retained images were subsequently augmented to obtain 900 samples for each co-existing disease combination.

3) Model Configuration and Training

To evaluate the effectiveness of the proposed synthetic dataset, four classification models were trained: **SVM**

TABLE I: Dataset Statistics I

Dataset Stage	BLB	Blast	Brown Spot
Before cleaning	2063	3178	2565
After cleaning	1076	2760	1392
Number of extracted lesions	495	452	145

TABLE II: Dataset Statistics II

Dataset Stage	BLB Blast	BLB Brown	Blast Brown
After Copy-Paste	1272	1086	1054
After Blending	329	322	342
After Augmentation	900	900	900

[20], **EfficientNet-B4** [23], **Swin Transformer** [22], and **ViT-Base/16** [21]. These models were selected to represent traditional ML, CNN, and transformer-based

architectures, respectively.

The SVM classifier [20] utilized deep feature embeddings extracted from frozen ResNet-50 backbone [24], whereas the remaining models were initialized with ImageNet pre-trained weights and fine tuned using transfer learning [16]. EfficientNet-B4 was trained using a two-stage transfer learning strategy by initially freezing the backbone and subsequently fine-tuning the upper layers. Both transformer models were optimized using the AdamW optimizer with the Binary Cross-Entropy with Logits(BCEWithLogits) loss function. Binary cross-entropy with logits loss was used because each disease label is treated as an independent binary prediction. EfficientNet-B4 was trained as a multi-label classifier using a sigmoid output layer, allowing multiple disease labels to be predicted from the same leaf image. Each image was represented using a three-dimensional binary label vector corresponding to BLB, Blast, and Brown Spot. For example, an image containing BLB and Blast was assigned the label vector [1, 1, 0] where each position independently indicates the presence or absence of the corresponding disease. The ViT-Base/16 model was similarly configured for multi-label disease classification and trained using the combined single-disease and synthetic co-occurrence data. All models were trained and evaluated using the same dataset partitions to ensure a fair comparison, and their performance was measured using Accuracy, Precision, Recall, and F1-score. Among the evaluated approaches, the transformer-based models provided stronger multi-label classification performance than the SVM and EfficientNet-based approaches.

4) Evaluation Metrics

The performance of the classification models was evaluated using four standard metrics: **Accuracy, Precision, Recall, and F1-score**. Accuracy measures the overall proportion of correctly classified samples. Precision evaluates the proportion of correctly predicted positive labels among all predicted positives. Recall measures the ability of the model to correctly identify the actual positive labels. The F1-score provides the harmonic mean of Precision and Recall, giving a balanced assessment of classification performance. These metrics were selected to evaluate the effectiveness of the generated synthetic dataset for multi-label rice leaf disease classification.

VI. RESULTS AND DISCUSSION

This section presents a qualitative and quantitative evaluation of the proposed synthetic dataset. The coexisting rice leaf disease images generated were assessed through visual inspection, expert validation, objective image quality metrics, and classification experiments to determine their realism and suitability for disease recognition tasks [12].

A. Qualitative Assessment of Synthesized Images

The generated images were first examined visually to determine whether the synthesized samples preserved the

appearance of a rice leaf while introducing the intended co-existing disease patterns. The proposed image-processing pipeline produced consistent composites across the three disease combinations, namely BLB+Blast, BLB+Brown Spot, and Blast+Brown Spot. The inserted lesions remained localized to the leaf region and the transition between the lesion and surrounding tissue was visually smooth. This is consistent with the use of feathered alpha compositing followed by multi-band Laplacian pyramid blending, which was designed to reduce visible boundaries while retaining the texture of the target leaf. Representative examples of these generated combinations are shown in Fig. 6

For comparison, three generative approaches, namely **CLIP-guided pixel optimization [30], cGAN [27][28] and BLIP-Diffusion [29]**, were also examined. The CLIP-guided outputs generally retained a leaf-like appearance, but noticeable fine-grained noise was present over large portions of the images. This made the outputs less suitable for direct use as synthetic training samples, even when the overall object structure appeared reasonable. The cGAN outputs generally had smoother textures and, in some cases, appeared more photorealistic than the CLIP outputs. However, the quality was not uniform across samples, with some images containing localized structural artifacts, blurred regions, or distortions in the leaf appearance.

The BLIP-Diffusion results showed the largest variation in semantic consistency. While some generated samples exhibited image-like textures, several outputs did not correspond to the intended rice-leaf disease combinations and instead produced visually unrelated content. Therefore, visual realism alone was not sufficient to establish that an image was useful for the proposed task. In contrast, the proposed blending approach maintained the expected rice-leaf structure and disease identity across the evaluated combinations, making its outputs more consistent for construction of a controlled multi-disease dataset. The results for all three approaches across the three disease combinations are shown in Fig. 7.

B. Image Quality Assessment

The visual quality of the synthesized images was further evaluated using objective image quality metrics, including PSNR, SSIM[25], NIQE, BRISQUE, and FID. These metrics were selected to assess structural similarity, perceptual quality, naturalness, and feature-level consistency between synthesized and reference images. The results are in **Table III**

Although the proposed blending pipeline did not achieve the lowest NIQE or BRISQUE scores, the comparison demonstrates that no-reference image-quality metrics alone are insufficient for evaluating the usefulness of synthetic disease images. As shown in **Table IV** BLIP-Diffusion[29] obtained the lowest NIQE score, while cGAN[27][28] achieved the lowest BRISQUE score; however, visual inspection revealed semantic inconsistencies and structural artifacts in their outputs. In contrast, the proposed pipeline consistently preserved the rice-leaf structure and generated the intended co-existing disease

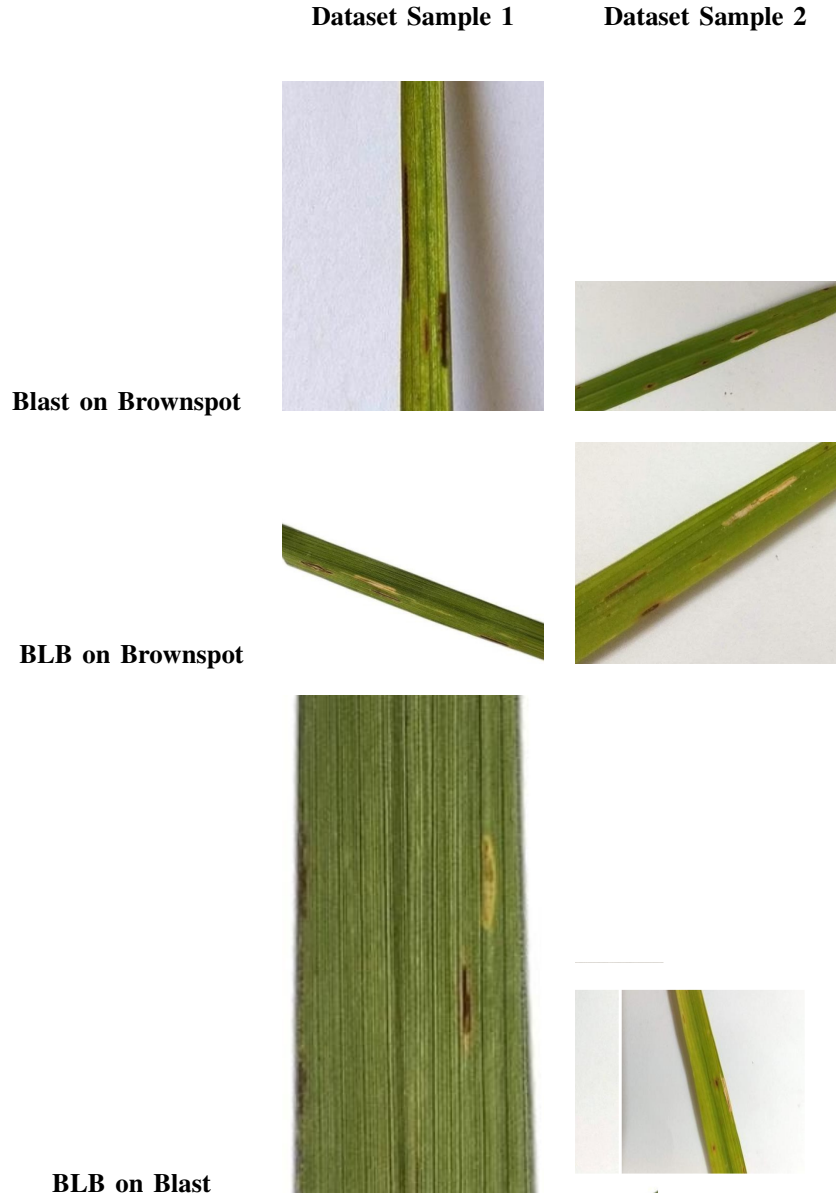


Fig. 4: Samples of final Dataset

TABLE III: Image Quality Assessment of the Proposed Synthetic Dataset

Metric	Value
PSNR	39.92
SSIM	0.965
NIQE	7.036
BRISQUE	45.471
FID	47.756

TABLE IV: Comparison of No-Reference Image Quality Metrics for Alternative Generative Approaches

Method	NIQE	BRISQUE
BLIP-Diffusion	6.541	41.851
cGAN	11.503	13.721
CLIP	9.085	59.592

patterns. Thus, for the present application, the consistency and controllability of disease synthesis were considered more important than optimizing a single general-purpose image-quality metric.

C. Classification Performance

To evaluate the effectiveness of the generated synthetic dataset for multi-label rice leaf disease classification, experiments were conducted using four representative models: **SVM[20]**, **EfficientNet-B4[23]**, **Swin Transformer[22]**, and **ViT-Base/16[21]**. The classification results are summarized (in percentage) in **Table V**. The classification results show a clear

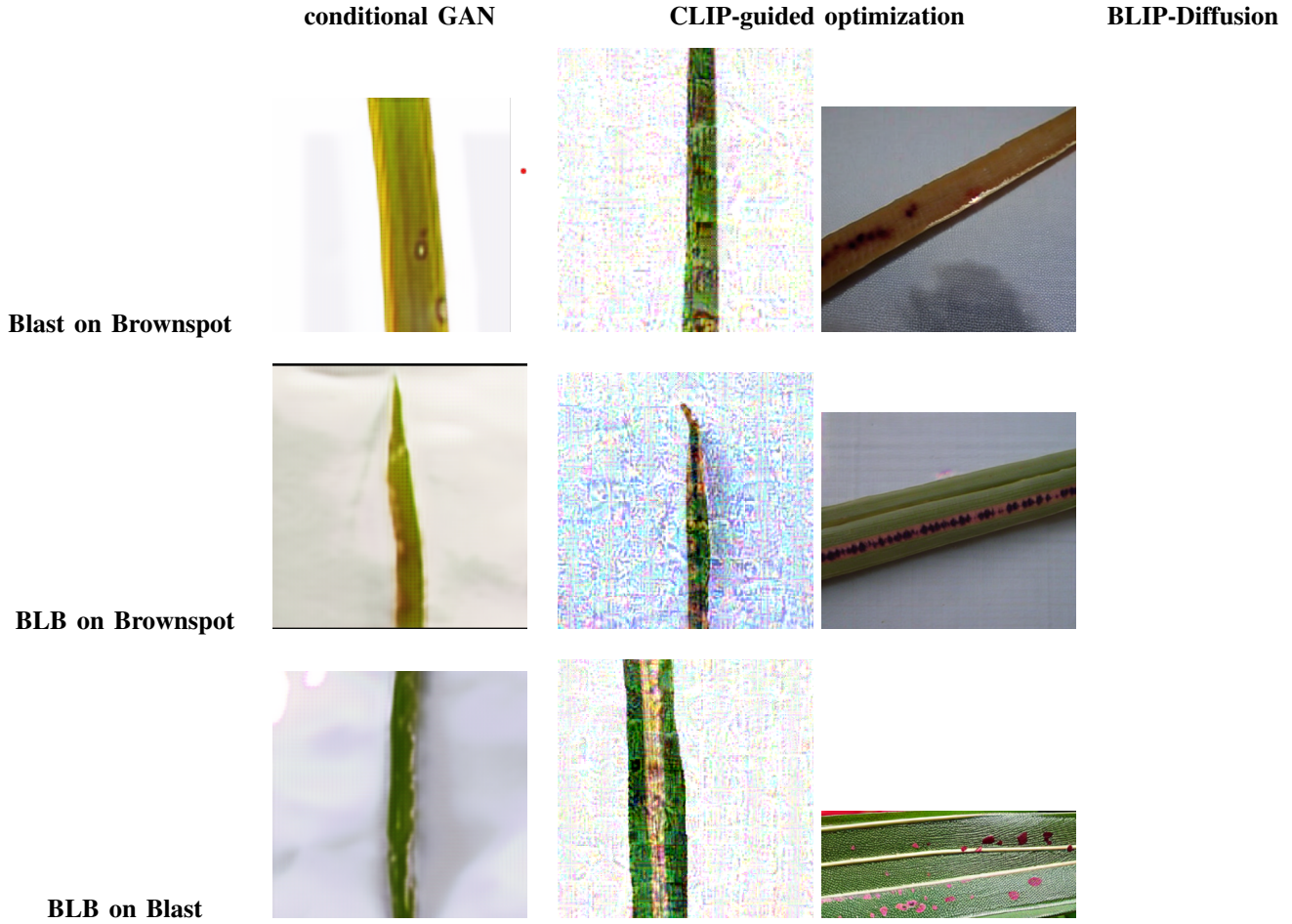


Fig. 5: Comparative visualization of three alternative generative approaches

TABLE V: Classification Performance of the Evaluated Models

Model	Accuracy	Precision	Recall	F1-score
SVM	50.25	100.00	50.00	66.67
EfficientNet-B4	66.70	72.00	48.00	57.6
ViT-Base/16	87.68	94.01	93.69	93.83
Swin Transformer	88.41	94.79	93.34	94.06

improvement with transformer-based models. SVM achieved an accuracy of 50.25%, while EfficientNet-B4[23] improved the accuracy to 66.70%. In comparison, ViT-Base/16 [21][17] and Swin Transformer[22] achieved substantially higher accuracies of 87.68% and 88.41%, respectively. Swin Transformer obtained the highest overall accuracy and F1-score of 94.06%, while ViT-Base/16 [21] achieved slightly higher recall (93.69%). These results indicate that transformer-based models were more effective in distinguishing the synthetic co-existing disease classes than the conventional SVM and EfficientNet-B4 models.

D. Discussion

The results indicate that the effectiveness of a synthetic image-generation method for agricultural datasets cannot be determined solely from general image-quality metrics. Although cGAN [27][28] achieved the lowest BRISQUE score and BLIP-Diffusion[29] obtained the lowest NIQE score among the evaluated generative approaches, visual inspection revealed inconsistencies such as artifacts, excessive noise, and semantically unrelated outputs. In contrast, the proposed image-processing approach provided direct control over the target leaf, lesion placement, and disease combination, resulting in more consistent synthetic samples. The classification experiments further showed that the generated dataset contained useful disease-related information, with ViT-Base/16[21] and Swin Transformer[22] achieving accuracies of 87.68% and 88.41%, respectively. Therefore, the proposed approach was selected for the final dataset because of its consistency and controllability rather than because it achieved the best score on every individual image-quality metric.

VII. LIMITATIONS

Despite these results, the proposed approach has limitations. Since the synthesis is based on transferring existing disease lesions between real leaf images, the diversity of generated samples is inherently dependent on the diversity of the available source images. The method may therefore provide less variation than fully generative approaches. Future work can investigate improved generative models with stronger disease-specific conditioning while retaining explicit control over disease combinations and lesion characteristics. For this specific controlled co-existing rice disease dataset-generation task, the deterministic approach was more reliable and controllable than the generative alternatives in this study. Another limitation of the present work is that the synthetic dataset is limited to pairwise co-existing disease combinations, namely BLB + Brown Spot, Blast + Brown Spot, and BLB + Blast. The current pipeline can be further extended to generate samples containing all three disease co-occurrences simultaneously, which would provide a broader representation of co-existing disease conditions in rice leaves. Such an extension could provide additional data for pathologists and farmers to study more complex disease interactions and support earlier identification and appropriate disease-management initiatives. Future work will therefore focus on extending the proposed framework to three-disease combinations and exploring a wider range of disease patterns and their possible interactions.

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